

The Inverted Stability Engine: A Formal Hybrid–Spectral Model of Bistable Contact Dynamics

Abstract

We formalize a common physical phenomenon: a long rigid body supported by two contacts can exhibit bistable “left/right” equilibria and will switch between them under external forcing (e.g. human pulling) due to a double-well effective energy landscape. We develop (i) a scalar double-well reduction with dissipation and a work-based switching criterion; (ii) an exact saddle-node (hysteresis) threshold for a biased quartic potential; and (iii) a graph–spectral formulation in which contact constraints are represented by a weighted Laplacian, including a connectivity-preserving hybrid “graph switching” lemma (“edge-bridge” switching) and a coupled network generalization (a spectral constraint engine). The resulting framework provides a rigorous bridge from practical mechanics to hybrid linear algebra and Laplacian eigenstructure.

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1 Problem Setup and Modeling Goals

1.1 Informal phenomenon

A rigid board resting on two lower supports can be observed to “prefer” one of two lateral contact configurations. When an external force is applied (pulling, rocking, or changing load), the board can abruptly switch from one configuration to the other, sometimes repeatedly, as if it were opposing the applied motion.

1.2 Formal goals

We aim to formalize:

- (G1) **Bistability:** conditions under which two locally stable equilibria exist (left/right).
- (G2) **Switching:** a sufficient, energy-based condition for a trajectory to cross the separatrix between wells.
- (G3) **Hysteresis threshold:** an exact saddle-node threshold under a bias parameter (tilt/unequal friction).
- (G4) **Spectral bridge:** a Laplacian formulation linking constraints to quadratic energy and eigenstructure.
- (G5) **Hybrid graph switching:** a lemma showing bistability persists under connectivity-preserving graph changes with bounded Laplacian perturbation and eigen-gap control.
- (G6) **Constraint engine:** a coupled network generalization with stability bounds and an Ising reduction.

2 Scalar Double-Well Reduction (Physical Core)

2.1 State variable

Definition 2.1 (Scalar lateral state). Let $x \in \mathbb{R}$ denote a signed lateral coordinate describing the board’s effective contact bias (e.g. signed displacement of the center-of-pressure projection or effective contact patch) relative to a symmetry axis defined by the supports. The states $x > 0$ and $x < 0$ represent two distinct contact configurations (“right” and “left”).

2.2 Dissipative forced dynamics

Definition 2.2 (Bistable board system). A *bistable board system* is modeled by

$$\ddot{x} + c\dot{x} + \partial_x V(x; \theta) = u(t), \quad (1)$$

where $c \geq 0$ is effective dissipation, $u(t)$ is an external generalized input, θ is a parameter vector (support spacing, geometry, friction, offsets), and $V(x; \theta)$ is an effective potential capturing gravity and contact constraints (after appropriate reduction).

Assumption 2.1 (Near-symmetry). In the symmetric baseline configuration, $V(x; \theta) = V(-x; \theta)$. Small asymmetries (tilt, unequal friction) appear as an odd perturbation in V .

2.3 Canonical quartic model

Definition 2.3 (Canonical biased double-well). We say the system is in the *canonical biased double-well regime* if

$$V(x; \theta) = ax^4 - bx^2 + \varepsilon x + r(x; \theta), \quad (2)$$

with $a(\theta) > 0$, $b(\theta) > 0$, bias $\varepsilon \in \mathbb{R}$, and remainder r of higher order (or sufficiently small) near equilibria.

2.4 Existence of two stable equilibria (unbiased)

Lemma 2.1 (Two stable equilibria, unbiased quartic). Assume $\varepsilon = 0$, $a > 0$, $b > 0$, and the remainder r is sufficiently small in a neighborhood of the quartic equilibria. Then the unforced dynamics

$$\ddot{x} + c\dot{x} + \partial_x V(x; \theta) = 0$$

admits two locally asymptotically stable equilibria near

$$x_{\pm} = \pm \sqrt{\frac{b}{2a}},$$

and an unstable equilibrium near $x = 0$.

Proof. For $r \equiv 0$, $\partial_x V = 4ax^3 - 2bx$ gives equilibria $x = 0$ and $x_{\pm} = \pm \sqrt{b/(2a)}$. The second derivative is $\partial_{xx} V(x) = 12ax^2 - 2b$, hence $\partial_{xx} V(0) = -2b < 0$ (unstable), and $\partial_{xx} V(x_{\pm}) = 12a(b/(2a)) - 2b = 4b > 0$ (local minima). With $c > 0$ the equilibria at minima are asymptotically stable by standard Lyapunov (energy) arguments. Small r perturbs equilibria continuously by the implicit function theorem. $\square$

2.5 Energy barrier and switching by work injection

Definition 2.4 (Barrier height). In the symmetric case ($\varepsilon = 0$), define the barrier height

$$\Delta V := V(0; \theta) - V(x_+; \theta) = V(0; \theta) - V(x_-; \theta). \quad (3)$$

Theorem 2.1 (Work-based sufficient switching condition). Consider (1) with $c \geq 0$. Let $E(t) = \frac{1}{2}\dot{x}(t)^2 + V(x(t); \theta)$. If, over an interval $[t_0, t_1]$, the injected work satisfies

$$\int_{t_0}^{t_1} u(t)\dot{x}(t) dt > \Delta V + \int_{t_0}^{t_1} c\dot{x}(t)^2 dt \quad (4)$$

and the trajectory begins in a neighborhood of one stable well, then the trajectory must cross the separatrix (the unstable manifold of the saddle near $x \approx 0$), enabling a switch to the opposite well.

Proof. Differentiate $E(t)$ along solutions:

$$\dot{E}(t) = \dot{x}\ddot{x} + \partial_x V(x)\dot{x} = \dot{x}(u - c\dot{x}) = u\dot{x} - c\dot{x}^2.$$

Integrate from t_0 to t_1 :

$$E(t_1) - E(t_0) = \int_{t_0}^{t_1} u\dot{x} \, dt - \int_{t_0}^{t_1} c\dot{x}^2 \, dt.$$

If the trajectory remains in the initial basin, $E(t)$ cannot exceed the saddle energy by at least ΔV (up to small modeling errors); inequality (4) forces $E(t_1)$ above the barrier, implying a crossing of the separatrix is necessary. $\square$

Remark 2.1. This theorem is the rigorous expression of “pulling injects enough work to hop the barrier, but friction dissipates some of it.” It does not require detailed geometry, only that the scalar reduction to an effective potential is valid near the observed dynamics.

3 Exact Saddle-Node Threshold for the Biased Quartic

This section gives the concrete, measurable threshold you can test in a physical rig by slowly introducing a bias (tilt, friction asymmetry, or support offset) and observing when one stable configuration disappears.

3.1 Stationary points and bifurcation condition

Consider the pure biased quartic with no remainder:

$$V(x) = ax^4 - bx^2 + \varepsilon x, \quad a > 0, \, b > 0. \quad (5)$$

Stationary points solve

$$V'(x) = 4ax^3 - 2bx + \varepsilon = 0. \quad (6)$$

A saddle-node bifurcation occurs when a stable equilibrium collides with the unstable one, i.e. when

$$V'(x) = 0 \quad \text{and} \quad V''(x) = 0 \quad (7)$$

simultaneously.

3.2 Closed-form critical bias

Theorem 3.1 (Exact saddle-node threshold). *For the biased quartic (5), the saddle-node condition (7) occurs at*

$$x_c = \pm \sqrt{\frac{b}{6a}} \quad (8)$$

and the corresponding critical bias magnitude is

$$|\varepsilon_c| = \frac{4}{3\sqrt{6}} \frac{b^{3/2}}{\sqrt{a}}. \quad (9)$$

For $|\varepsilon| < |\varepsilon_c|$, (6) has three real roots (two minima and one maximum), producing bistability; for $|\varepsilon| > |\varepsilon_c|$, there is a unique real root and bistability vanishes.

Proof. Compute $V''(x) = 12ax^2 - 2b$. Setting $V''(x) = 0$ gives $x^2 = b/(6a)$, hence (8). Plug into $V'(x) = 4ax^3 - 2bx + \varepsilon$:

$$\varepsilon = 2bx - 4ax^3 = 2bx \left(1 - 2ax^2/b\right).$$

Using $x^2 = b/(6a)$ yields $2ax^2/b = 2a(b/(6a))/b = 1/3$, so

$$\varepsilon = 2bx \left(1 - \frac{1}{3}\right) = \frac{4}{3}bx.$$

Thus $|\varepsilon_c| = \frac{4}{3}b|x_c| = \frac{4}{3}b\sqrt{\frac{b}{6a}} = \frac{4}{3\sqrt{6}}\frac{b^{3/2}}{\sqrt{a}}$. The root-count statement follows from standard cubic discriminant analysis.

Corollary 3.1 (Hysteresis under quasi-static bias). *If ε is varied slowly in time, the stable equilibrium on one side persists until $|\varepsilon|$ reaches $|\varepsilon_c|$, at which point it disappears in a saddle-node bifurcation and the state jumps to the remaining well. Reversing ε produces the opposite jump, yielding hysteresis in the mapping $\varepsilon \mapsto \text{sign}(x)$.*

Remark 3.1. In the physical system, ε subsumes any net odd bias: tilt, unequal contact friction, support height mismatch, or torque bias from applied forces. The threshold (9) is the clean “hard number” for when one basin collapses.

4 Spectral Contact Model: Graph Laplacian as Constraint Energy

We now formalize the bridge to spectral graph structure: contact constraints induce a quadratic energy expressed by a Laplacian; nonlinearities introduce double-well stabilization.

4.1 Graph primitives

Definition 4.1 (Weighted contact graph). Let $G = (V, E, w)$ be a weighted undirected graph with $|V| = n$. The weighted adjacency matrix $A \in \mathbb{R}^{n \times n}$ satisfies $A_{ij} = w_{ij}$ if $(i, j) \in E$ and $A_{ij} = 0$ otherwise. The degree matrix is $D = \text{diag}(d_1, \dots, d_n)$ with $d_i = \sum_j A_{ij}$. The Laplacian is

$$L := D - A. \tag{10}$$

Definition 4.2 (Dirichlet (graph) energy). For a node variable $z \in \mathbb{R}^n$ (generalized displacement/contact coordinate), define

$$\mathcal{E}_G(z) := \frac{1}{2}z^\top Lz = \frac{1}{2} \sum_{(i,j) \in E} w_{ij}(z_i - z_j)^2. \tag{11}$$

Remark 4.1. $\mathcal{E}_G$ is the canonical quadratic penalty for disagreement along edges. In contact mechanics terms: it is a stiffness-like energy that grows when coupled contact variables separate.

4.2 Connectivity and algebraic connectivity

Definition 4.3 (Algebraic connectivity). Let $0 = \lambda_1(L) \leq \lambda_2(L) \leq \dots \leq \lambda_n(L)$ be the eigenvalues of L . The quantity $\lambda_2(L)$ is the *algebraic connectivity*.

Lemma 4.1 (Connected iff $\text{Ker}(L) = \text{span}\{\mathbf{1}\}$). *A weighted undirected graph G is connected if and only if*

$$\text{Ker}(L) = \text{span}\{\mathbf{1}\},$$

equivalently $\lambda_2(L) > 0$.

4.3 Nonlinear contact: quartic stabilization

Pure Laplacian energy (11) is quadratic and cannot by itself create a double-well. The observed bistability arises from *nonlinear contact/friction geometry*.

Definition 4.4 (Quartic-regularized spectral energy). Let $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a stabilizing quartic term, e.g.

$$\Phi(z) = \frac{\alpha}{4} \sum_{i=1}^n z_i^4, \quad \alpha > 0. \quad (12)$$

Define the total energy

$$\mathcal{H}_G(z) = \frac{1}{2} z^\top L z - \frac{\beta}{2} \|z\|^2 + \Phi(z) - h^\top z, \quad (13)$$

where $\beta > 0$ encodes an effective destabilizing curvature (from geometry or load transfer), and h encodes bias/forcing in generalized coordinates.

Remark 4.2. (13) is the minimal spectral analogue of the scalar quartic: quadratic coupling ($z^\top L z$) plus a negative isotropic quadratic ($-\beta \|z\|^2$) creates an instability; the quartic regularizer confines the state and produces multi-well structure.

5 Hybrid Linear Algebra: Graph-Space Switching with Edge-Bridges

This section formalizes your core intuition in the language of hybrid linear algebra: *switch the graph (constraint topology) while preserving connectivity and key spectral structure*.

5.1 Hybrid graph family

Definition 5.1 (Graph family and mode index). Let $\{G_k\}_{k \in \mathcal{K}}$ be a family of weighted graphs on a common vertex set V , with Laplacians L_k and energies $\mathcal{H}_{G_k}$ as in (13). A *hybrid graph trajectory* is a pair $(z(t), \sigma(t))$ with

$$\sigma(t) \in \mathcal{K} \text{ piecewise-constant}, \quad \dot{z} = -\nabla \mathcal{H}_{G_{\sigma(t)}}(z) + \eta(t),$$

where $\eta(t)$ is external input/noise/actuation (including human intervention).

Definition 5.2 (Connectivity-preserving switch). A switch $k \rightarrow \ell$ is *connectivity-preserving* if both graphs are connected, i.e.

$$\text{Ker}(L_k) = \text{Ker}(L_\ell) = \text{span}\{\mathbf{1}\} \iff \lambda_2(L_k) > 0, \lambda_2(L_\ell) > 0.$$

5.2 Laplacian perturbation and spectral continuity

Lemma 5.1 (Weyl bound for Laplacian eigenvalues). For symmetric matrices L_k, L_ℓ ,

$$|\lambda_i(L_k) - \lambda_i(L_\ell)| \leq \|L_k - L_\ell\|_2 \quad \forall i.$$

Lemma 5.2 (Subspace stability under eigen-gaps (Davis–Kahan principle)). If $\lambda_2(L_k)$ is separated from $\lambda_3(L_k)$ by a gap $\gamma_k := \lambda_3(L_k) - \lambda_2(L_k) > 0$, then the second-eigenspace varies continuously under perturbations $L_\ell = L_k + \Delta$ with $\|\Delta\|_2 < \gamma_k$, with a quantitative bound on the principal angle proportional to $\|\Delta\|_2 / \gamma_k$.

5.3 Edge-bridge switching lemma (your “connectome bridge”)

Definition 5.3 (Edge-bridge switch). An *edge-bridge switch* is a change $G_k \rightarrow G_\ell$ that modifies only a bounded subset of edges/weights (adding/removing/rewiring a limited number of constraints), so that $\Delta L := L_\ell - L_k$ has bounded operator norm $\|\Delta L\|_2 \leq \delta$.

Theorem 5.1 (Connectome Edge-Bridge Lemma: structural bistability under graph switching). *Assume:*

- (A1) (**Connectivity preserved**) G_k and G_ℓ are connected, hence $\text{Ker}(L_k) = \text{Ker}(L_\ell) = \text{span}\{\mathbf{1}\}$.
- (A2) (**Bounded bridge**) $\|L_\ell - L_k\|_2 \leq \delta$.
- (A3) (**Spectral safety margin**) $\lambda_2(L_k) \geq \underline{\lambda} > 0$ and $\lambda_2(L_\ell) \geq \underline{\lambda} > 0$ (no connectivity collapse).
- (A4) (**Low-mode bistable reduction**) There exists a 1D reduced coordinate $x = \langle z, \psi \rangle$ (for some unit vector $\psi \perp \mathbf{1}$) such that, restricted to an invariant or attracting manifold $\mathcal{M}$, both energies $\mathcal{H}_{G_k}$ and $\mathcal{H}_{G_\ell}$ induce effective scalar potentials $V_k(x)$ and $V_\ell(x)$ in the canonical biased double-well regime (2), with parameters $(a_k, b_k, \varepsilon_k)$ and $(a_\ell, b_\ell, \varepsilon_\ell)$ satisfying $a_k, a_\ell > 0$ and $b_k, b_\ell > 0$.

Then:

- (C1) The existence of two metastable basins (left/right) is structurally stable under the switch $k \rightarrow \ell$ provided $|\varepsilon_k| < |\varepsilon_{c,k}|$ and $|\varepsilon_\ell| < |\varepsilon_{c,\ell}|$, where each critical bias is given by (9) with $(a, b) = (a_k, b_k)$ or (a_ℓ, b_ℓ) .
- (C2) If δ is small and the eigen-gap near the low-mode manifold is maintained, then the reduced coordinate direction ψ (interpretable as a “soft” mode / second-eigenvector-like direction) varies continuously across the switch, hence the meaning of “left vs right” is preserved up to a controlled rotation.

Proof sketch. (A1) ensures the nullspace remains exactly $\text{span}\{\mathbf{1}\}$, so global connectivity is preserved. (A2) plus Weyl’s inequality gives bounded eigenvalue drift, in particular keeping λ_2 away from 0 under (A3). Subspace stability (Davis–Kahan principle) controls rotation of the low-mode direction as long as an eigen-gap remains. Finally, bistability is an unfolding-stable property of the double-well catastrophe class: as long as each mode stays in the regime $|\varepsilon| < |\varepsilon_c|$ (with ε_c given by (9)), the two minima persist under perturbations. $\square$

Remark 5.1. This theorem is the formal statement of: “switch graph space while holding connectivity, and the bistable attractor structure persists.” The key technical levers are (i) λ_2 bounded away from 0, (ii) bounded Laplacian perturbation, and (iii) remaining in the same catastrophe class.

6 Spectral Constraint Engine: Coupled Network Generalization

We now extend from a single bistable element to a *network* of bistable elements coupled by a graph. This is the mechanical analogue of a constraint solver: topology defines couplings; nonlinear wells define discrete-like states; the system relaxes toward consistent configurations.

6.1 Network model

Definition 6.1 (Spectral constraint engine). Let $G = (V, E, w)$ be a connected graph with Laplacian L . Let $x \in \mathbb{R}^n$ be node states. Define the energy

$$H(x) := \sum_{i=1}^n \left(\frac{\alpha}{4} x_i^4 - \frac{\beta}{2} x_i^2 \right) + \frac{\kappa}{2} x^\top L x - h^\top x, \quad (14)$$

with parameters $\alpha, \beta, \kappa > 0$ and bias $h \in \mathbb{R}^n$. Define dissipative dynamics

$$\dot{x} = -\nabla H(x) + \xi(t), \quad (15)$$

where $\xi(t)$ is an optional perturbation/noise/input.

Remark 6.1. The term $\frac{\kappa}{2} x^\top L x = \frac{\kappa}{2} \sum_{(i,j)} w_{ij} (x_i - x_j)^2$ couples nodes to agree along edges. The local terms create bistability in each coordinate. This is the minimal “spectral constraint engine” consistent with the scalar phenomenon.

6.2 Existence of bounded equilibria

Lemma 6.1 (Coercivity and boundedness of minimizers). *If $\alpha > 0$ then $H(x) \rightarrow +\infty$ as $\|x\| \rightarrow \infty$. Hence H admits at least one global minimizer and all sublevel sets $\{x : H(x) \leq c\}$ are bounded.*

Proof. The quartic term grows as $\sum_i \alpha x_i^4/4$, dominating the negative quadratic terms and linear bias. The coupling term is nonnegative. Thus $H(x)$ is coercive and achieves a minimum on $\mathbb{R}^n$. $\square$

6.3 Stability near equilibria via Hessian bounds

Let $\nabla^2 H(x)$ denote the Hessian:

$$\nabla^2 H(x) = \text{diag}(3\alpha x_i^2 - \beta) + \kappa L.$$

Theorem 6.1 (Local stability condition at an equilibrium). *Let x^* be an equilibrium of (15) (i.e. $\nabla H(x^*) = 0$) with $\xi \equiv 0$. If $\nabla^2 H(x^*)$ is positive definite, then x^* is locally asymptotically stable for (15).*

Proof. Linearizing (15) yields $\dot{\delta} = -\nabla^2 H(x^*)\delta$. If the Hessian is positive definite, all eigenvalues of $-\nabla^2 H(x^*)$ are strictly negative, so the equilibrium is exponentially stable. $\square$

Proposition 6.1 (A conservative PD sufficient condition). *If for all i ,*

$$3\alpha(x_i^*)^2 - \beta \geq \mu \quad \text{with} \quad \mu > 0,$$

then $\nabla^2 H(x^) \succeq \mu I$ (since $\kappa L \succeq 0$), hence x^* is locally asymptotically stable.*

Remark 6.2. This sufficient condition is conservative: it guarantees stability if each coordinate sits sufficiently deep in its well. Coupling via κL can stabilize even if some coordinates are shallower.

6.4 Reduction to Ising (discrete spin) approximation

Definition 6.2 (Spin reduction). Assume each coordinate is near one of two well minima and define $s_i \in \{-1, +1\}$ by

$$s_i := \text{sign}(x_i).$$

In the deep-well regime, approximate $x_i \approx ms_i$ where $m > 0$ is a characteristic amplitude (e.g. $m \approx \sqrt{\beta/\alpha}$ when coupling is weak).

Theorem 6.2 (Ising approximation of the spectral constraint engine). *Under the approximation $x \approx ms$ with $s \in \{-1, +1\}^n$, the energy (14) reduces (up to an additive constant) to an Ising-type objective*

$$H(s) \approx - \sum_{(i,j) \in E} J_{ij} s_i s_j - \sum_{i=1}^n \tilde{h}_i s_i, \quad (16)$$

with couplings $J_{ij} = \kappa w_{ij} m^2$ and fields $\tilde{h}_i = h_i m$.

Proof. Compute $x^\top Lx = \sum_{(i,j)} w_{ij} (x_i - x_j)^2 = 2 \sum_{(i,j)} w_{ij} (x_i^2 + x_j^2 - 2x_i x_j)$. The x_i^2 terms contribute constants under $x_i^2 \approx m^2$. The cross terms yield $+2\kappa \sum_{(i,j)} w_{ij} (-x_i x_j)$ which becomes $-2\kappa m^2 \sum_{(i,j)} w_{ij} s_i s_j$, absorbing constants into the offset. The linear term $-h^\top x$ becomes $-\sum_i h_i m s_i$. $\square$

Remark 6.3. This is the clean “network surprise”: your lumber bistability is a single spin; a coupled lattice is an analog optimizer. Topology $\leftrightarrow$ constraint graph; dynamics $\leftrightarrow$ energy descent.

7 Synthesis: How the Three Pieces Lock Together

- S1. Scalar physical core:** the observed behavior is captured by a dissipative forced double-well model. Switching occurs when injected work exceeds barrier plus dissipation.
- S2. Exact thresholds:** under a bias parameter ε , bistability persists exactly for $|\varepsilon| < \frac{4}{3\sqrt{6}} \frac{b^{3/2}}{\sqrt{a}}$. Past this, one basin collapses (saddle-node).
- S3. Spectral/hybrid bridge:** constraints and couplings are Laplacian-quadratic; nonlinear contact creates quartic stabilization; changing the graph (edge-bridge switching) preserves bistability if connectivity and eigen-gap remain protected and the reduced mode stays in the same catastrophe class.
- S4. Constraint engine:** coupled bistables on a graph produce a spectral constraint engine whose deep-well limit reduces to an Ising objective, connecting physical structure to discrete optimization.

8 Practical Measurement Targets (for a physical rig)

Remark 8.1. Although the document is theoretical, it yields directly testable quantities.

- (M1) Measure the *bias* at which one stable configuration disappears (tilt angle, friction differential, or support offset). Fit the observed threshold to (9) to estimate $b^{3/2}/\sqrt{a}$.
- (M2) Measure switching under controlled forcing: estimate injected work $\int u \dot{x} dt$ and compare to barrier plus dissipation, per (4).

- (M3) For a coupled rig (multiple bistable elements), estimate effective couplings via $J_{ij} \approx \kappa w_{ij} m^2$ and compare observed stable patterns to low-energy Ising states (16).

9 Closing Note

This formalization isolates the essential structure: bistability is a low-dimensional catastrophe (double-well) induced by geometry and stabilized by nonlinearity; spectral graph structure provides the natural language for constraint coupling; and hybrid graph switching is stable so long as connectivity and spectral safety margins are preserved.